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Local Coverage Governs Memorization in Diffusion Models

A Sharp Phase Transition Separates Memorized from Generalised Outputs—and It Depends Only on Neighbourhood Density

By Claudia Merger & Sebastian Goldt (SISSA / ICTP, Trieste)

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A new theoretical result explains why diffusion models memorise some training examples and generalise on others—sometimes simultaneously. The criterion is purely local: a training point is memorised if and only if its neighbourhood in the data space contains too few other training samples for the model to interpolate. In the high-dimensional limit, the transition is sharp and its critical threshold scales as $n^{-1/k}$ with dataset size n and intrinsic data dimensionality k .

AT A GLANCE

Problem: No principled theory predicts which training points a diffusion model will memorise versus generalise.

Why it matters: Memorisation enables training-data extraction attacks, raises copyright concerns, and reveals a gap in our theoretical understanding of diffusion generative models.

Method: Exploit the KDE interpretation of the diffusion score function to derive a closed-form local coverage criterion for memorisation, with a sharp phase transition in the high- d limit.

Result: The criterion correctly predicts memorised vs. novel samples on CIFAR-10 and CelebA without any model-specific information.

Evidence: Validated on synthetic Gaussian mixtures ($d \in \{10, 50, 200\}$), CIFAR-10, and CelebA.

The Big Picture

Diffusion models generate images by iteratively denoising a Gaussian noise vector. When trained on a large dataset, most outputs are novel; but a small fraction—particularly samples close to rare or isolated training images—are effectively verbatim copies of training data. This memorisation is not a bug in the model; it is a fundamental property of how the score function behaves near sparse regions of the data distribution.

The community has documented memorisation empirically (cataloguing which prompts or seeds trigger it) and proposed detection heuristics (membership inference, attention-map analysis). What has been missing is a simple, principled answer to the question: *why does a particular training point get memorised?*

This paper answers that question. Leveraging the well-known connection between score matching and kernel density estimation (KDE), the authors show that the score function is dominated by the nearest training neighbours of any query point. If a training point has too few neighbours within the relevant kernel bandwidth, the score field around it points straight back to that point at every noise level—ensuring the reverse diffusion ODE converges there. In high dimensions, this transition is sharp.

Why Existing Approaches Fall Short

Empirical approaches study memorisation as a global property of the trained model: “this model memorises 3% of its training set.” They cannot predict *which* 3% without running the model.

Attention-map analyses (Bright Ending Attention, BEA) detect memorisation post hoc by examining the model’s internal activations. They require model access and do not provide a generalisable theory.

Membership inference attacks frame memorisation as a binary classification problem, but they neither explain the mechanism nor provide a quantitative criterion depending on data geometry.

Core Mechanism

The score function of a diffusion model at noise level σ is well-approximated (for overparameterised networks) by the KDE score:

$$\hat{s}(x, \sigma) = \nabla_x \log \hat{p}_\sigma(x), \quad \hat{p}_\sigma(x) = \frac{1}{n} \sum_{i=1}^n \mathcal{N}(x; x_i, \sigma^2 I).$$

Near a training point x_j , the score is:

$$\hat{s}(x_j, \sigma) \approx \frac{\sum_{i \neq j} (x_i - x_j) w_{ij}(\sigma)}{1 + \sum_{i \neq j} w_{ij}(\sigma)}, \quad w_{ij}(\sigma) = e^{-\|x_i - x_j\|^2 / 2\sigma^2}.$$

When $\sum_{i \neq j} w_{ij}(\sigma) \ll 1$ (no close neighbours), the score at x_j points approximately to zero—i.e., toward x_j itself at all scales. The reverse ODE then converges to x_j from any initialisation in its vicinity: memorisation.

Technical Formulation

Memorisation criterion. Define the *local coverage* of x_j at scale σ :

$$C_\sigma(x_j) = \frac{1}{n} \sum_{i \neq j} \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right).$$

Training point x_j is **memorised** iff $C_\sigma(x_j) \ll 1/n$ for all $\sigma \in [\sigma_{\min}, \sigma_{\max}]$ (the range of noise levels used during training).

Phase transition in high dimensions. For data supported on a k -dimensional manifold of radius R , the number of training samples within ball $B(x_j, \sigma)$ is $\bar{n}(\sigma) \approx n(\sigma/R)^k$. Memorisation occurs iff $\bar{n}(\sigma) = O(1)$, i.e., iff $\sigma < \sigma^*$ where:

$$\sigma^* \sim R n^{-1/k}.$$

This is the **critical radius**: points x_j with no neighbours within σ^* are always memorised; points with many neighbours within σ^* always generalise. The transition is *sharp* in the limit $d \rightarrow \infty$, $n \rightarrow \infty$ with k fixed.

Predictor. The paper proposes a practical memorisation predictor: compute $C_{\hat{\sigma}^*}(x_j)$ in the embedding space of a pre-trained feature extractor and threshold at $C^* = 1/n$. This requires no model access—only the dataset.

Learning or Inference Procedure

The analysis is purely theoretical; no new training procedure is introduced. The practical predictor works as follows:

1. Embed all training points with a feature extractor ϕ .
2. Estimate the intrinsic dimension k of the training set via nearest-neighbour methods.
3. Compute $\sigma^* = R n^{-1/k}$ where R is the dataset diameter.
4. For each training point x_j , compute $C_{\sigma^*}(x_j)$.
5. Predict memorisation if $C_{\sigma^*}(x_j) < 1/n$.

What the Guarantee Says

In the high-dimensional limit ($d \rightarrow \infty$ with $k/d \rightarrow 0$), the transition is shown to be sharp: for any $\varepsilon > 0$, there exist constants $c_1, c_2 > 0$ such that

$$\Pr[\text{memorised} \mid C_\sigma(x_j) < c_1/n] \rightarrow 1,$$

$$\Pr[\text{generalised} \mid C_\sigma(x_j) > c_2/n] \rightarrow 1,$$

as $n, d \rightarrow \infty$. Memorisation probability varies continuously between these extremes at finite n .

Experimental Findings

- **Synthetic:** On Gaussian mixtures with controlled inter-point distance, the empirical memorisation fraction matches the theoretical prediction to within measurement noise at $d \in \{10, 50, 200\}$.
- **CIFAR-10:** The local coverage criterion identifies memorised samples (confirmed by running inference from fixed seeds) with $> 85\%$ precision and $> 80\%$ recall without any model access.
- **CelebA:** Rare attribute combinations (e.g., red hair + glasses) are correctly predicted as memorised; common attributes (e.g., brown eyes) are correctly predicted as generalised.

Ablations and Interpretation

Feature space: Using raw pixel distances yields weaker predictions than CLIP embeddings (Δ -precision $\approx +15\%$), confirming that perceptual distance is the relevant metric.

Kernel bandwidth: Results are robust to σ within a factor of 2 of σ^* , consistent with the sharp transition theory.

Dataset size: As n increases, the fraction of memorised samples decreases as $n^{-1/k}$, matching the theoretical scaling.

Reference

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